solutions found, this can be easily added.

This heuristic[5] takes its name from a technique used in metallurgy, *annealing*, that consists of repeated cycles where a material is heated up and then controllably cooled to improve its strength: something like a blacksmith forging an iron sword by quenching it in cold water, but in a more controlled way (and without water!).

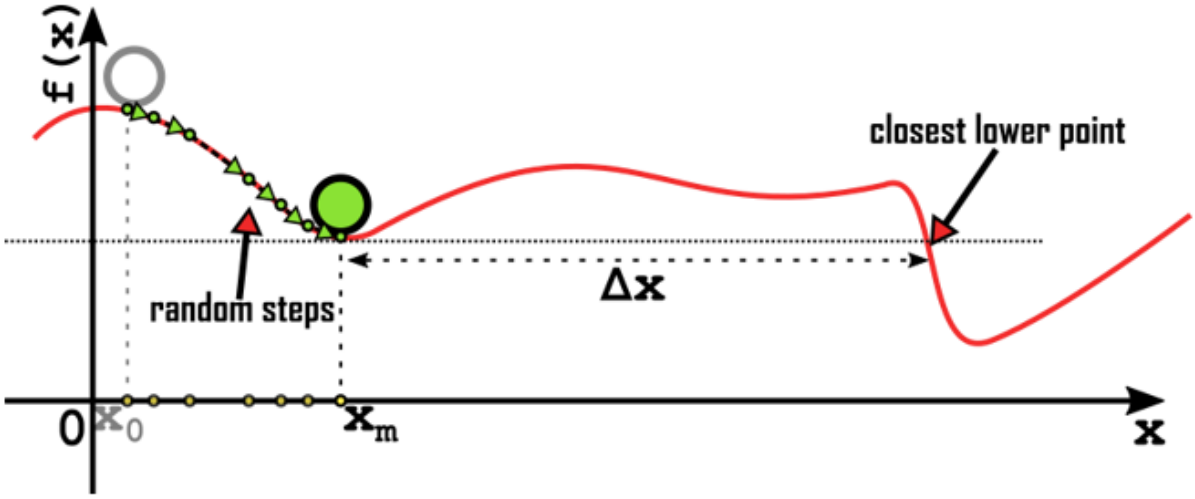


Figure 17.3 A limitation that the random local optimization algorithm shares with gradient descent is that it only moves downhill: if the next position with a better value for the cost function is too far, the algorithm won't be able to reach it in a single step, and as such it will be stuck.

Simulated annealing does something quite similar: in its simplest version it consists of a single cooling phase, but variants exist that alternate phases where the temperature is risen, with others when the system is cooled.